Response to the Second-Round Review

BayesianDEB: A Bayesian Framework for Dynamic Energy Budget Modelling in R

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on behalf of the authors

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We are grateful to the editor and reviewer for reading the manuscript so closely a second time. Each comment from the review of **22 May 2026** is quoted below in italics, with our reply underneath and the files we touched to address it. Everything here refers to version **0.2.1** of the package; the replication archive is `BayesianDEB_replication.zip` and the revised manuscript is `bayesiandeb.tex`.

In short:

- a single master replication script (`replicate_all.R`) rebuilds every manuscript number in about a minute from cached draws;
- every code listing in the manuscript now runs, and prints its output, in the replication scripts;
- all manuscript outputs come from the shipped cache under a fixed seed, so they reproduce bit for bit;
- the diagnostics `print` and `plot` methods are now compact;
- the `bdeb_diagnose` examples have moved from `\dontrun{}` to a `cmdstanr`-gated `\donttest{}` block;
- the dose-response figure now draws the reference lines its caption describes;
- the README installation instructions no longer pull the stale CRAN version.

1 Package version must match the CRAN version

> "The submitted package version is not in line with the package published on CRAN. We require that both versions match."

Response. When the reviewer wrote, CRAN still held 0.1.4 while the manuscript and this submission describe 0.2.1. That gap is now closed: 0.2.1 went up on CRAN on 2026-06-17, so the published package and the manuscript describe the same release. The version bump is documented in `cran-comments.md`, and `DESCRIPTION` declares `Version: 0.2.1`. `R CMD check --as-cran` reports 0 errors, 0 warnings and a single note (the informational `cmdstanr-in-Additional_repositories` line, discussed below).

2 README should include cmdstanr installation instructions

> "The README file should also include instructions for the installation of cmdstanr."

Response. Added. The Installation section of `README.md` now spells out the two steps: first install `cmdstanr` from the Stan r-universe, then build the CmdStan toolchain once,

```
install.packages("cmdstanr",
  repos = c("https://stan-dev.r-universe.dev", getOption("repos")))
cmdstanr::install_cmdstan()
```

and the text notes that `bdeb_fit()` checks for the toolchain at run time.

3 A single replication script that runs within one hour

> "Journal of Statistical Software requires that a single replication script that can be run within one hour is provided. ... parts of the results require that the scripts in `sbc` are run and these scripts are supposed to take more than 20 hours to finish. We would thus ask that the authors arrange their replication scripts into a smaller number of R files and that they also work on reducing the computational time needed to replicate results."

Response. There is now a single master script, `replication/replicate_all.R`, that rebuilds every figure, table and printed number in the manuscript from one command (`Rscript replicate_all.R`). Out of the box it runs no MCMC at all: it reads the archived posterior draws in `outputs/` and `sbc/` and regenerates everything in roughly a minute on a laptop, comfortably under the one-hour limit. The lengthy simulation-based-calibration runs stay out of this path. Their rank matrices are shipped (`sbc/*.rds`) and reproduce Figures 3–4 and Tables 4–5 on their own; recomputing them from scratch takes anywhere from 45 min to 12 h per model and is described separately in `sbc/README.md`, outside the replication budget. Anyone who does want to refit can set `BDEB_RECOMPUTE=true BDEB_MODE=full` for a complete run ($\sim$3 h) or `BDEB_MODE=lite` for a quick one ($\sim$30 min).

Reference. `replication/replicate_all.R`, `replication/README.md`, `replication/sbc/README.md`.

4 `example("bdeb_diagnose")` should be executable (`\donttest`)

> "`example("bdeb_diagnose")` shows that this example is in `\dontrun`. This should, in principle, be executable code and we expect it to be in `donttest` instead so as to allow its execution using `example`, similarly to what has been implemented for `bdeb_fit`."

Response. Done. The examples for `bdeb_diagnose()`, the `bdeb_diagnostics` `print`/`summary`/`plot` methods, `bdeb_loo()`, `bdeb_derived()` and the deprecated `bdeb_summary()` are now wrapped in the same cmdstanr-gated `\donttest{}` block we already use for `bdeb_fit()`:

```
\donttest{
if (requireNamespace("cmdstanr", quietly = TRUE) &&
    nzchar(tryCatch(cmdstanr::cmdstan_path(), error = function(e) ""))) {
  data(eisenia_growth)
  dat <- bdeb_data(growth = eisenia_growth[eisenia_growth$id == 1, ])
  fit <- bdeb_fit(bdeb_model(dat, type = "individual"),
                  chains = 2, iter_warmup = 200, iter_sampling = 200,
                  refresh = 0)
  print(bdeb_diagnose(fit))
}
}
```

so `example("bdeb_diagnose")` runs whenever a CmdStan toolchain is available. None of the exported functions rely on `\dontrun{}` any more.

Reference. R/diagnostics.R, R/fit.R, R/ppc.R, R/debtox.R.

5 The plot method for bdeb_diagnostics could be more readable

> "We believe that the plot method for bdeb_diagnostics could be improved to allow a shorter and more readable display on screen."

Response. plot.bdeb_diagnostics() no longer crowds the display with per-time-point latent states (x_sol[i,j], L_hat[i], and so on). By default it shows only the scalar model parameters, so the R-hat and ESS panels stay short and easy to read rather than running to dozens of latent rows. Passing full = TRUE brings back the complete plot, and the subtitle notes how many latent rows were left out. This follows what print.bdeb_diagnostics() already does.

Reference. R/diagnostics.R (plot.bdeb_diagnostics); NEWS.md.

6 Replication output differs from the article (LOO comparison)

> "Results obtained with the replication material (full version) are still different from the results shown in the article. For instance, on page 13, the loo_compare for the lognormal model differs."

Response. This comparison really is sensitive to Monte-Carlo noise on this individual. With only $n = 13$ observations, a handful of Pareto-$\hat{k}$ values climb above 0.7, so the importance-sampling ELPD difference moves by a fraction of its standard error from run to run; across our reruns it landed around -1 every time, always with SE ≈ 1.3 and always within one standard error of zero. We made three changes:

- fixed a global seed (seed = 42) in every fit, so the shipped result is deterministic;
- had replicate_all.R reproduce the manuscript numbers from the cached draws, so the printed ΔELPD matches the article exactly;
- rewrote the conclusion as a qualitative one: since $|\Delta\text{ELPD}|$ is smaller than its standard error, neither observation model wins clearly, and the precise difference is best read as indicative.

The manuscript prints lognormal -1.0 (SE 1.3) and says outright that the difference sits within its standard error.

Reference. replication/00_setup.R (seed), replication/01_illustrations.R, manuscript Section 5.1.

7 README installation pulls the old CRAN version (0.1.4)

> "The instructions in the README.md suggest installing with a repos option that includes CRAN, which installs version 0.1.4 and not the submitted package version 0.2.0."

Response. We have split the installation instructions. BayesianDEB is now installed with a plain install.packages("BayesianDEB"), which now resolves to 0.2.1 from CRAN, while cmdstanr is installed separately from the Stan r-universe (see Section 2). The earlier combined repos = c("https://stan-dev.r-universe.dev", getOption("repos")) call, which could have masked the CRAN copy of BayesianDEB, is gone from the README.

Reference. README.md (Installation section).

8 Replication material is incomplete (missing manuscript code)

> "Some code included in the manuscript which should be executable code is not contained ... E.g., on page 7, `prior_species("Daphnia_magna", type = "debtox")`."

Response. Every code listing in the manuscript is now run in the replication scripts. `prior_species("Daphnia_magna", type = "debtox")`, for example, is called and printed in `replication/01_illustrations.R` next to the `prior_species("Eisenia_fetida")` listing. We went back through the manuscript one listing at a time and checked that each appears in a replication script.

Reference. `replication/01_illustrations.R`.

9 `bdeb_diagnose(fit)` output differs and is too long

> "Some results obtained differ. E.g., compared to page 12, `bdeb_diagnose(fit)` reports a different number of divergent transitions and prints a very long table including `x_sol[i,j]` and `L_hat[i]` rows."

Response. Two things address this:

- **Reproducibility.** With the fixed seed and cached draws (Section 6), `replicate_all.R` reproduces the page-12 output exactly; the manuscript now reports a single divergent transition (1 of 4000 draws, $< 0.1\%$) and comments on it.
- **Length.** `print.bdeb_diagnostics()` now defaults to the scalar model parameters and hides the per-time-point latent states, so the p. 12 output is compact (eight parameter rows, no `x_sol`/`L_hat`). The full table remains available through `print(x, full = TRUE)` or `summary(x)$table`.

Reference. `R/diagnostics.R`, `replication/01_illustrations.R`, manuscript Section 5.1.

10 `bdeb_derived()` result is created but never printed

> "The replication material creates `der <- bdeb_derived(fit, ...)` but does not print it, so it is not easy to compare with the manuscript."

Response. Fixed. `replication/01_illustrations.R` now prints the derived-quantity table right after it is built, laid out as in the manuscript:

```
der <- bdeb_derived(fit,
  quantities = c("L_m", "L_inf", "k_M", "g", "growth_rate"), f = 1.0)
print(as.data.frame(summarise_draws(der, "mean", "sd",
  "q5" = ~quantile(.x, 0.05), "q95" = ~quantile(.x, 0.95))), digits = 3)
```

Reference. `replication/01_illustrations.R`.

11 LOO triggers Pareto-$\hat{k}$ warnings that are not commented on

> "Warnings are triggered (`Some Pareto k diagnostic values are too high`) where one might also point them out and comment on them."

Response. These warnings are now flagged and explained rather than left to surprise the reader. In `replication/01_illustrations.R` a comment ahead of the `loo_compare()` call points out that, at $n = 13$, a few $\hat{k} > 0.7$ are to be expected and the importance-sampling ELPD is therefore only indicative. The manuscript makes the same point in the text of Section 5.1: it reports the

Pareto-$\hat{k}$ warning, explains what it means (the approximation is unreliable at a few points), and uses it to justify reading the comparison qualitatively.

Reference. `replication/01_illustrations.R`, manuscript Section 5.1.

12 Figure caption refers to lines that are not in the figure

> "The caption of Figure 2 discusses something dashed red and dotted green. However, it is unclear to what in the figure one refers."

Response. `plot_dose_response()` now draws the lines the caption promises: a dashed red vertical line at the posterior-median EC_{50} and a dotted green one at the posterior-median NEC, with a matching legend subtitle. Non-finite draws are discarded and the view is clipped with `coord_cartesian()`, so degenerate draws no longer leave vertical artefacts. The Figure 2 caption and the rendered figure now agree.

Reference. `R/debtox.R` (`plot_dose_response`), manuscript Figure 2 caption.

13 Unclear which part of the replication reproduces Figures 3 and 4

> "It is unclear what part of the replication material reproduces Figures 3 and 4."

Response. `replication/README.md` now carries a *Figure* → *script* table that ties each manuscript figure to the script and cached object behind it. For the two in question: Figure 3 (`fig_sbc_ranks.pdf`) comes from `02_validation.R` via `sbc/sbc_results.rds`, and Figure 4 (`fig_sbc_hierarchical.pdf`) comes from `02_validation.R` via `sbc/sbc_hierarchical_results.rds`. The same table covers the remaining figures and the matching manuscript sections.

Reference. `replication/README.md` (Figure → script and Section → script tables).

We thank the editor and reviewer once more for feedback that has made the package and its replication material better.

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